SCEPI: Single-Mutant-Conditioned Epistasis Prediction for Compensatory Protein Variant Discovery

Dong Xu[1,2,4], Jiayu Zhang[2], Zhangfan Yang[3], Jun Zhang[1,4], Zexuan Zhu[1,4], and Junkai Ji[1,2,4⋆]

[1] School of Artificial Intelligence, Shenzhen University, 518060 Shenzhen, China
[2] EasternDawn, Shenzhen, China
[3] School of Computer Science, University of Nottingham Ningbo, 315100 Ningbo, China
[4] Intelligent Computing Center, Shenzhen University, 518060 Shenzhen, China

Abstract. Selecting compensatory double mutants requires predicting how individually damaging substitutions combine. We introduce Single-mutant-Conditioned Epistasis Prediction (SCEPI), a cross-protein model that conditions a shared interaction function on measured single-mutant effects. These measurements also calibrate individual effects outside the interaction network. Source combination measurements train the model, while target double-mutant labels remain withheld. Across 155 held-out proteins, SCEPI achieves mean residual root mean squared error of 0.595 kcal/mol, compared with 0.805 for ThermoMPNN-D, 0.847 for Kermut, and 0.853 for calibrated StabilityArc. At a budget of ten candidates per protein, its compensation hit rate is 11.22% across 120 proteins, compared with 7.07%, 5.40%, and 8.75%, respectively. This rate is 4.27 times the 2.63% expectation of uniform random selection from the same candidates. Matched input-removal controls retain external calibration and establish prediction and selection gains from measured single effects within the interaction network. Common measured main effects, stricter homology grouping, and measurement-boundary analysis preserve these gains. SCEPI uses existing single-mutant measurements to guide compensatory discovery in unseen proteins.

Keywords: Protein stability · Epistasis · Compensatory mutations · Deep mutational scanning · Double-mutant selection

1 Introduction

Compensatory substitutions recover stability or function lost through other sequence changes [1, 2, 5]. Their discovery requires selecting useful combinations from individually damaging substitutions. Measured single-mutant effects provide target-specific information for directing a fixed experimental budget toward those combinations.

Complete wild-type, single-mutant, and double-mutant measurements define observed nonadditivity [12]. Source proteins supply this supervision for learning a shared interaction function. At inference, target single-mutant measurements condition predictions for unmeasured combinations. Global phenotype nonlinearity and pair-dependent effects both contribute to the observed contrast [9, 10].

Transferable combination predictors learn from structure or sequence representations [3, 8, 13]. Target-specific regression also uses measured variants [6]. These capabilities motivate a question about the additional value of supplying target single-mutant measurements directly to a shared interaction function.

We introduce SCEPI, which conditions a shared interaction function on available single-mutant effects and their observation indicators. The same measurements calibrate individual effects outside that function. The pair function combines these inputs with reference and single-mutant representations. Its four-state contrast vanishes at either wild-type endpoint and cancels endpoint-only offsets. Source combination measurements train the shared parameters.

Matched input-removal controls retain external calibration, model capacity, training samples, and selection procedures. Removing the measured pair inputs increases residual error and reduces compensation hit

⋆ Corresponding author.

rate under both homology groupings. The full-target comparison gives SCEPI a top-10 hit rate of 11.22%, compared with 7.07% for ThermoMPNN-D, 5.40% for Kermut, and 8.75% for calibrated StabilityArc. Every method receives zero target double-mutant training labels. Common measured main effects and measurement-boundary analysis preserve the selection gains. A same-capacity direct-head comparison tests the four-state reference constraints.

2 Related work

Combination predictors from protein representations. ThermoMPNN-D uses a Siamese head over frozen ProteinMPNN features [3]. Mutate Everything composes reusable mutation representations [8]. ESM-MSR encodes wild-type and mutant sequences [11]. The StabilityArc preprint learns a shared wild-type decoder and a symmetric, contact-conditioned residual [13]. These models establish transferable combination prediction from protein representations.

Using target measurements. Table 1 distinguishes the roles of single-mutant measurements. Mutate Everything constructs training residuals from measured singles and uses predicted effects at inference [8]. StabilityArc supplies a cross-protein prior to Kermut's target Gaussian-process posterior [6, 13]. SCEPI supplies measured effects directly to its shared pair network and external calibration. The matched input-removal control isolates the pair-input role.

Phenotype maps and interaction terms. MAVE-NN, MoCHI, and reference-free analysis fit genetic effects and nonlinear phenotype maps [4, 10, 14]. Global epistasis separates phenotype nonlinearity from pair-dependent contributions [9]. SCEPI retains both in its predicted phenotype contrast. Component comparisons evaluate the map and fitting objectives separately from measured-input conditioning.

3 Methods

SCEPI combines a source-trained interaction function with measured target single effects (Figure 1). These measurements enter the pair network and external calibration. Target double-mutant outcomes are reserved for evaluation.

3.1 Prediction problem

Let s_p be a reference protein and M a set of substitutions at distinct sites. The predictor receives s_p, M, and the target measurements allowed by the information condition. It outputs the relative phenotype $\widehat{\mu}_p(M)$ and a candidate score. For two substitutions $A = (i, a)$ and $B = (j, b)$, the measured interaction is

$$\epsilon_p(A, B) = y_p(AB) - y_p(A) - y_p(B) + y_p(\varnothing). \tag{1}$$

Table 1. Single-mutant information use. The final column records direct measured-value inputs to the pair network. ThermoMPNN-D target-head fitting refers to our single-only target protocol. Mutate Everything is included for the methodological comparison.

Method	Source use of single labels	Target-single use	Measured pair inputs
ThermoMPNN-D [3]	Over-and-back augmentation	Target-head fitting	No
Mutate Everything [8]	Single supervision and residual targets	Predicted singles at inference	No
StabilityArc + Kermut [6, 13]	Decoder supervision	Target GP posterior	No
SCEPI	Joint phenotype and residual supervision	External calibration and pair conditioning	Effects and availability

The four observations share a reference background, assay condition, and phenotype scale. We call them a complete quartet. Stability is expressed as unfolding free energy, with positive changes indicating increased stability. Target single-mutant labels are available in the main compensation task. Target test double-mutant labels are used to evaluate predictions and selected candidates.

3.2 Mutation representations and calibrated single effects

We use a frozen ESM-2 encoder h [7]. For substitution $A = (i, a)$, the endpoint representation contains $h_i(s_p)$, the local change $h_i(x_p(A)) - h_i(s_p)$, amino-acid identities, and the pretrained substitution score. A shared head predicts the single effect through a reference subtraction,

$$a_i(a) = f_\phi(u_i(a)) - f_\phi(u_i(w_i)), \tag{2}$$

where w_i is the reference residue. Thus $a_i(w_i) = 0$. The same head processes every protein.

Available single-mutant effects enter through

$$\widetilde{a}_i(a) = a_i(a) + \omega m_i(a)\big[\mu_i^{\mathrm{obs}}(a) - a_i(a)\big], \tag{3}$$

where $m_i(a)$ records label availability. The mixing weight ω is selected on source-validation proteins. A measured zero remains an available label. The calibrated effects enter both the single-effect sum and the interaction network.

Pair representations also contain mutation-dependent cross-site context and sequence separation. Appendix A defines these features and the structural variant.

3.3 Conditioning the shared interaction function

For a candidate pair, collect the calibrated single effects and their availability indicators in

$$c_{ij}(a, b) = \big(\widetilde{a}_i(a), \widetilde{a}_j(b), m_i(a), m_j(b)\big). \tag{4}$$

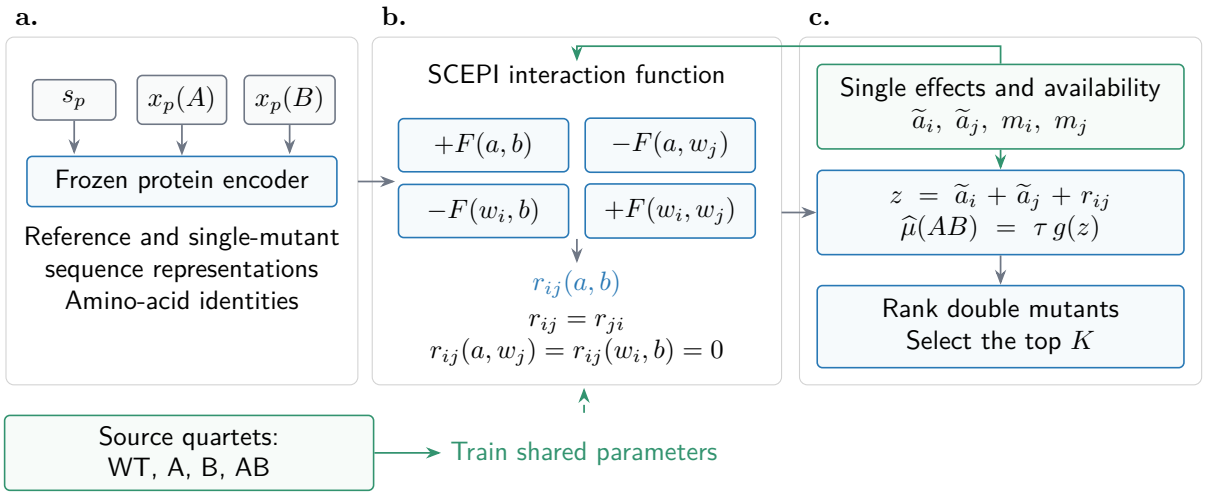


Fig. 1. SCEPI conditions shared interaction predictions on single-mutant measurements. (a) A frozen encoder supplies reference and mutation features from the reference sequence and its single mutants. (b) Calibrated single effects and their availability indicators condition every evaluation of the shared symmetric pair function. Its four-state contrast defines the pair output and enforces reference-state constraints. (c) The same single effects enter the external main-effect sum before phenotype prediction. Double-mutant predictions then rank candidates at a fixed budget. Complete source quartets and source observations supervise SCEPI. Target double-mutant outcomes are reserved for evaluation.

The shared network receives this vector together with the endpoint representations and site relation. Thus the pair function depends on the target's measured single effects while its parameters remain shared across proteins. Exchanging the endpoints also exchanges the entries of c_{ij}.

Let $\Psi_\theta(i, a; j, b \mid c_{ij})$ denote the ordered network. We average both orders:

$$F_\theta(i, a; j, b) = \tfrac{1}{2}\big[\Psi_\theta(i, a; j, b \mid c_{ij}) + \Psi_\theta(j, b; i, a \mid c_{ji})\big]. \tag{5}$$

The conditioning inputs enter every evaluation of F. We parameterize the pair output by its four-state contrast,

$$r_{ij}(a, b) = F(i, a; j, b) - F(i, a; j, w_j) - F(i, w_i; j, b) + F(i, w_i; j, w_j). \tag{6}$$

Every call uses the specified amino-acid state. Restoring an endpoint to wild type sets its effect and local representation change to zero. It also removes the context change caused by that endpoint at the other site.

Equation (6) gives symmetry, $r_{ij}(a, b) = r_{ji}(b, a)$, and zero reference effects, $r_{ij}(a, w_j) = r_{ij}(w_i, b) = 0$. Replacing F by $F + v_i(a) + v_j(b) + c$ leaves r unchanged because the four terms cancel every endpoint-only contribution. A matched direct-head comparison tests the predictive contribution of these constraints.

3.4 Composition and phenotype prediction

For a mutation set M, the latent effect sums single and pair contributions:

$$z_p(M) = \sum_{(i,a)\in M} \widetilde{a}_i(a) + \sum_{\substack{(i,a),(j,b)\in M \\ i<j}} r_{ij}(a, b). \tag{7}$$

We predict $\widehat{\mu}_p(M) = \tau g(z_p(M))$, where $\tau > 0$, $g(0) = 0$, and $g'(0) = 1$. The monotone map models global phenotype nonlinearity. Its parameterization is given in the appendix.

The model's interaction on the observed phenotype scale is always calculated from its four predictions:

$$\widehat{\epsilon}_p(A, B) = \widehat{\mu}_p(AB) - \widehat{\mu}_p(A) - \widehat{\mu}_p(B). \tag{8}$$

This contrast includes global phenotype nonlinearity and the learned pair contribution. It equals r_{ij} under the identity map and unit scale. Appendix A gives the decomposition.

3.5 Learning from complete combination measurements

The single-effect head is initialized from source single-mutant measurements. Source combination measurements then jointly fit the single head, pair function, and phenotype map. Training combines a Huber loss on the complete predicted residual with the likelihood of source observations. Sampling balances proteins and site pairs. The likelihood retains point-valued and interval-valued measurements.

Source training varies the permitted single-mutant information to represent inference conditions. Source-validation proteins determine model configurations and the label-mixing weight. Target homology groups remain excluded from fitting and selection. Appendix C gives the full criterion, observation model, and implementation. Residual-only and phenotype-only variants compare objective choices.

3.6 Compensatory candidate selection

SCEPI ranks double mutants using its native phenotype prediction $\widehat{\mu}_p(AB)$. For a fixed budget K, it selects the highest-scoring K candidates. We evaluate recovery relative to the better constituent single mutant,

$$R_p(A, B) = y_p(AB) - \max\{y_p(A), y_p(B)\}. \tag{9}$$

A compensatory hit satisfies $R_p(A, B) \geq \delta$. The candidate task focuses on pairs whose two measured single effects are each at most $-\delta$.

We also compare interaction information with a common measured main effect. Each method m ranks the same candidates by

$$s_m(A, B) = \mu_p^{\text{obs}}(A) + \mu_p^{\text{obs}}(B) + \widehat{\epsilon}_{m,p}(A, B). \tag{10}$$

Every method retains its own four-prediction contrast. This comparison fixes the single-effect sum and changes the predicted nonadditivity. Native ranking measures the complete method's utility. Equation (10) tests the contribution of interaction predictions to that utility.

Uniform random selection samples K candidates without replacement from the same protein-level pool. Its expected hit rate is the pool's hit fraction, and its expected mean recovery is the pool mean. Enrichment is the ratio of the model and random equal-protein mean hit rates. Appendix K gives the exact calculation and uncertainty analysis.

4 Experimental setting

We use the cDNA-display proteolysis stability measurements of Tsuboyama et al. [12]. The accepted double-mutant set contains 182,663 variants across 190 reference backgrounds. Its complete, point-valued quartets number 127,892 across 160 backgrounds and 488 site pairs. The retrospective evaluation cohort requires all four measurements to lie inside the assay range of $[-1, 5]$ kcal/mol. Model training also retains interval-valued observations through the observation likelihood.

The primary evaluation holds out homology groups and scores all accepted target double mutants. Grouping uses 30% sequence identity and at least 80% bidirectional coverage. Related reference backgrounds remain together, and annotated family relations further merge groups. Source training and validation exclude the outer test partition. S-all supplies all permitted target single-mutant labels and excludes every target double-mutant label from fitting. Homology groups and annotated families have separate coverage records.

Comparators include independently trained additive and global nonlinear models, ThermoMPNN-D, Kermut, StabilityArc + Kermut, mean ridge, and mean MLP. The global model retains the monotone map and omits the pair term. The mean-representation controls fit ridge regression or a multilayer perceptron (MLP) to averaged mutation representations. Source-trained baselines are retrained on the same source partitions, with checkpoints selected by source validation. StabilityArc scores supply the prior mean for Kermut target calibration [6, 13]. All methods receive zero target double-mutant labels and score the same candidates. Appendices C and D give their architectures, source supervision, target adaptation, and structural inputs.

Phenotype evaluation requires at least 30 point-valued target measurements per protein. It covers 159 proteins and 51 homology groups. Residual evaluation requires at least 30 complete, in-range quartets and covers 155 proteins from the same 51 groups. Compensation selection requires at least 100 eligible candidates per protein and reference stability at most 4.5 kcal/mol. Its cohort contains 120 proteins and 44 groups. We use candidate budgets $K \in \{1, 3, 10, 30\}$ and recovery threshold $\delta = 0.5$ kcal/mol. Metrics are calculated within each protein, averaged over matched repetitions, and then averaged equally across proteins. Hyperparameters and label-mixing weights are chosen using source-validation measurements.

The quartet boundary margin is the smallest distance of any endpoint from either assay limit. Margins of 0.25 and 0.5 kcal/mol retain 120 and 115 compensation proteins; the latter covers 43 groups. The designated site-pair subset and target combination adaptation have separate evaluations. Statistical and repetition details appear in the appendix.

Additional information conditions measure the value of target labels. Z permits no target measurements. S32 and S128 permit the reference and the specified number of target single-mutant labels. The C conditions additionally permit 128, 512, 2,048, or all available training double-mutant labels. Their designated site-pair holdout excludes test pairs from target combination fitting. Separate adaptation experiments cover six functional assays and nine FLIP2 split protocols [16]. Appendix I defines their measurement units, model variants, and evaluation populations.

5 Results

5.1 Measured single effects improve interaction prediction and selection

Conditioning the pair function on measured single effects improves prediction and compensation selection under both homology groupings (Table 2; Figure 4c,d). The matched input-removal control supplies model-predicted single effects and zero availability indicators to F. It retains the external calibration formula

Table 2. Matched pair-function comparisons under 30% and 25% homology grouping. Residual prediction covers 155 proteins, and native top-10 selection covers 120. The predicted-singles control removes measured values and availability indicators from F while retaining external calibration. The direct head keeps those inputs and replaces the four-state contrast with the same network's direct output.

	Residual RMSE (kcal/mol)		Native hit rate (%)	
Pair function	30%	25%	30%	25%
SCEPI	**0.595**	**0.604**	**11.22**	11.15
Predicted singles in F	0.767	0.814	8.18	8.15
Direct residual head	0.635	0.630	10.70	**11.38**

and source-validation selection procedure. Model capacity, initialization, training samples, objectives, and stopping rules are matched. Each model is refitted independently.

Measured pair inputs reduce residual RMSE by 22.4% under the primary grouping and 25.7% under the stricter grouping. Native top-10 hit rates improve by 3.03 and 3.00 percentage points. Both mean gains are positive in every matched training repetition under each grouping. The residual and compensation cohorts contain the same 155 and 120 proteins across controls.

The input contribution remains when every method uses a common measured main-effect sum. Hit-rate gains over the input-removal control are 3.08 and 3.02 percentage points under the two groupings. At the 0.5 kcal/mol quartet boundary margin, the gains are 3.11 and 3.13 points. These comparisons identify selection information supplied by measured inputs within the interaction function.

5.2 The conditional predictor improves compensatory candidate discovery

SCEPI achieves a mean top-10 compensation hit rate of 11.22% using its native predictions (Table 3). ThermoMPNN-D achieves 7.07%, Kermut 5.40%, and the global nonlinear model 4.45%. SCEPI improves their hit rates by 4.15, 5.82, and 6.77 percentage points. Its mean selected recovery exceeds these methods by 0.183, 0.201, and 0.180 kcal/mol. Recovery is higher in 86 of 120 proteins against ThermoMPNN-D and in 93 of 120 against Kermut. Figure 2 shows the full budget curves and all protein-level hit-rate differences.

The additive model reaches a native hit rate of 6.38%. SCEPI raises this rate by 4.83 percentage points and improves mean recovery by 0.130 kcal/mol. Mean MLP and mean ridge reach 10.07% and 11.42%. SCEPI has the highest mean selected recovery among the seven baselines at every evaluated budget. At $K = 10$, its recovery advantage over mean ridge is 0.098 kcal/mol. Native ranking therefore distinguishes successful discoveries from the continuous quality of the selected candidates.

StabilityArc + Kermut achieves an 8.75% top-10 compensation hit rate. SCEPI improves this rate by 2.47 percentage points and increases mean recovery by 0.087 kcal/mol. Mean recovery is higher in 73 of 120 proteins. This comparison evaluates SCEPI against a second source-trained architecture with explicit pair effects and target single-mutant calibration.

SCEPI enriches compensatory hits 4.27-fold over uniform random selection at $K = 10$. The same 120 proteins contain 79,564 eligible candidates. Random selection has an expected equal-protein hit rate of 2.63% and mean recovery of -0.900 kcal/mol. SCEPI increases the hit rate by 8.59 percentage points and mean recovery by 0.643 kcal/mol. Enrichment is 5.52, 4.90, 4.27, and 3.26 at budgets 1, 3, 10, and 30, respectively (Figure 2a,b).

5.3 The learned function predicts interactions in unseen combinations

Mean residual RMSE is 0.595 kcal/mol for SCEPI, 0.805 for ThermoMPNN-D, and 0.847 for Kermut. The respective error reductions are 26.1% and 29.8%. SCEPI has lower protein-level residual RMSE in 136 of 155 proteins against either method (Figure 3a).

StabilityArc + Kermut has a residual RMSE of 0.853 kcal/mol. SCEPI reduces this error by 30.3% and improves residual prediction in 130 of 155 proteins. Both methods learn pair effects across source proteins and use the permitted target single labels.

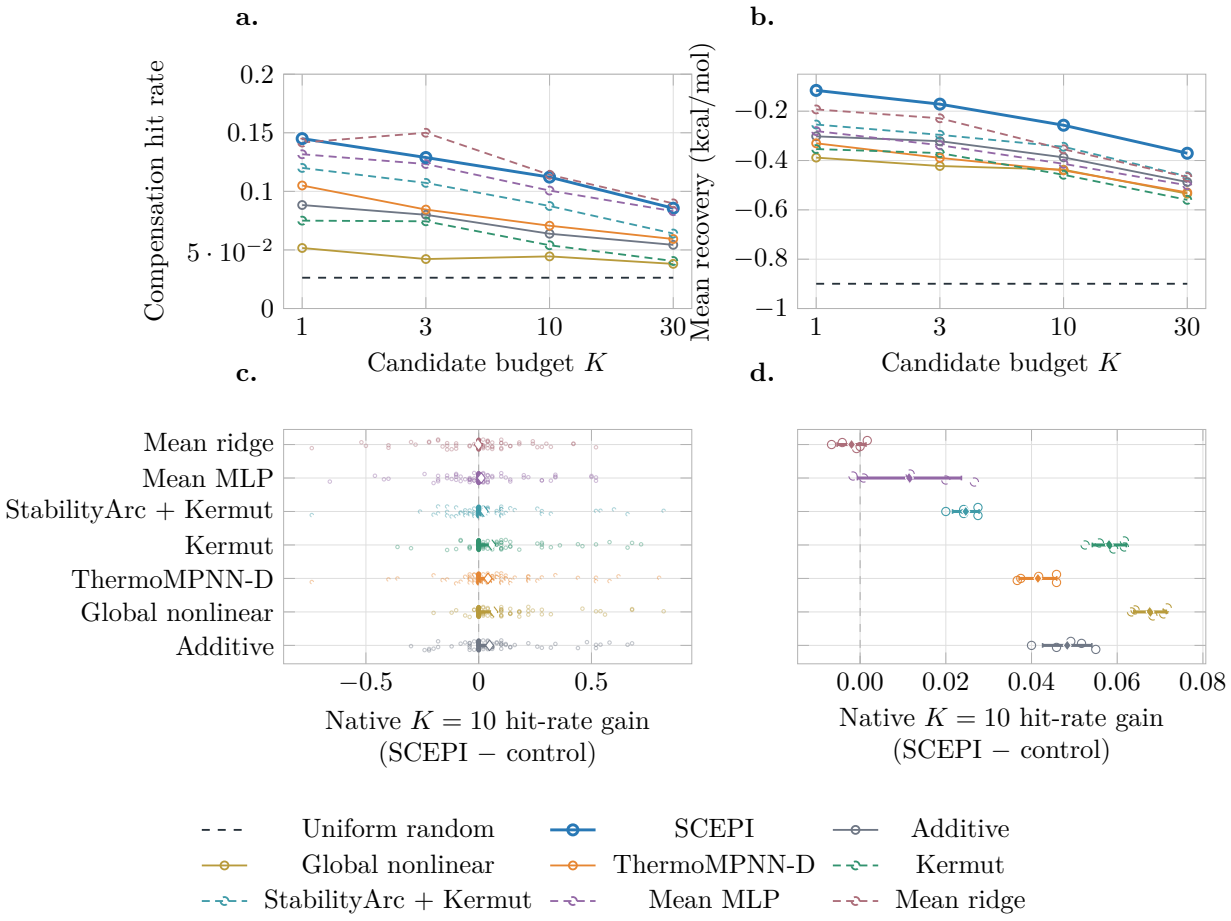


Fig. 2. SCEPI compensatory discovery in full-target S-all evaluation. (a) Native hit rate and (b) mean recovery across candidate budgets. Dashed black lines show exact uniform-random expectations from the same candidates. (c) Protein-level top-10 hit-rate gains after averaging repetitions within each of 120 proteins. Open points denote proteins; segments, ticks, and diamonds denote interquartile ranges, medians, and means. (d) Repetition-level gains in the same control order. Open points are equal-protein means; diamonds and whiskers show their mean and standard deviation. All predictors use identical candidates and native phenotype scores. StabilityArc + Kermut uses source-held-out scores with target calibration.

The global nonlinear control reaches 0.635 kcal/mol, and the additive control reaches 1.091 kcal/mol. Their comparison separates the contribution of a learned phenotype map from the additional pair function. SCEPI reduces the global control's mean residual error by 6.4%. Its error is lower in 92 of 155 proteins. The complete per-protein comparison retains both directions of difference.

Native phenotype prediction provides a separate test of SCEPI. Its phenotype RMSE is 0.657 kcal/mol and its mean Spearman correlation is 0.742. ThermoMPNN-D reaches 0.777 and 0.673, while Kermut reaches 0.798 and 0.696. StabilityArc + Kermut reaches 0.815 and 0.746. SCEPI has lower phenotype RMSE in 117 of 159 proteins against ThermoMPNN-D and in 114 of 159 against Kermut. Its mean phenotype and residual prediction errors also improve over both target-fitted mean-representation baselines. Figure 3b–d shows the protein-level differences and repetition-level residual gains.

With 128 target double-mutant training labels, the source-initialized SCEPI adaptation variant reaches unseen-site-pair RMSE of 0.567 kcal/mol across 70 proteins. Kermut reaches 0.758, ThermoMPNN-D 0.775, second-order MoCHI 0.859, and the target-only SCEPI adaptation variant 0.799. Table 13 reports every comparator across the four budgets.

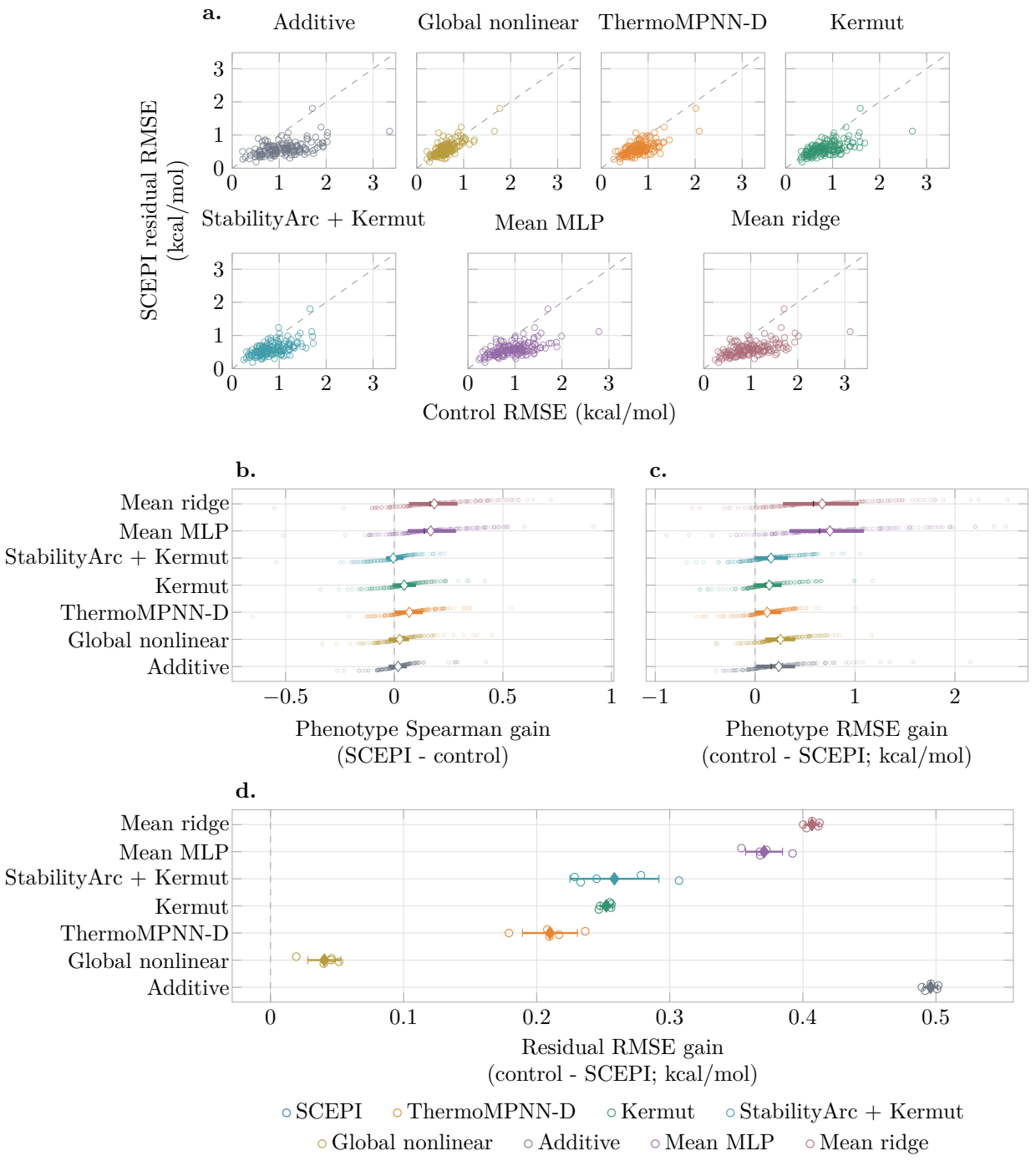


Fig. 3. SCEPI prediction across held-out homology groups in the full-target S-all setting. (a) Paired residual RMSE for all 155 eligible proteins. The dashed diagonal denotes equal error. (b) Native phenotype Spearman gain across 159 proteins. (c) Phenotype RMSE gain in the same cohort. In (b) and (c), points are proteins, segments are interquartile ranges, dark ticks are medians, and diamonds are means. (d) Repetition-level residual RMSE gains. Open points are equal-protein macro averages; diamonds and whiskers show the mean and standard deviation. Positive gains favor SCEPI. All eligible proteins are retained.

Table 3. Full-target prediction and native top-10 compensation selection in held-out homology groups. Phenotype, residual, and selection cohorts contain 159, 155, and 120 proteins. RMSE uses kcal/mol. Values are equal-protein means over matched repetitions. Enrichment is relative to uniform random selection from the same candidates. All models receive zero target double-mutant training labels.

Predictor	Phenotype RMSE	Residual RMSE	Hit rate (%)	Recovery (kcal/mol)	Enrichment (×)
SCEPI	**0.657**	**0.595**	11.22	−0.257	4.27
ThermoMPNN-D	0.777	0.805	7.07	−0.440	2.69
Kermut	0.798	0.847	5.40	−0.458	2.05
StabilityArc + Kermut	0.815	0.853	8.75	−0.344	3.33
Global nonlinear	0.911	0.635	4.45	−0.437	1.69
Additive	0.893	1.091	6.38	−0.387	2.43
Mean MLP	1.405	0.966	10.07	−0.413	3.83
Mean ridge	1.328	1.002	**11.42**	−0.354	**4.34**
Uniform random	—	—	2.63	−0.900	1.00

5.4 Interaction predictions add value beyond measured single effects

Fixing the main-effect sum across methods preserves the selection advantage (Figure 4a; Appendix Table 7). SCEPI achieves a compensation hit rate of 11.27%, compared with 4.42% for additivity and 4.70% for global nonlinearity. ThermoMPNN-D reaches 5.30%, Kermut reaches 5.47%, and StabilityArc + Kermut reaches 8.17%. Mean MLP and mean ridge reach 7.17% and 7.33%. Mean selected recovery improves by 0.176, 0.161, 0.224, and 0.200 kcal/mol over the first four controls. Its improvement over StabilityArc + Kermut is 0.095 kcal/mol. These paired scores use an identical measured single-effect sum. Their differences come from each method's predicted four-term nonadditivity.

At a quartet margin of at least 0.5 kcal/mol, SCEPI's hit rate is 11.15%. Additivity reaches 4.52%, global nonlinearity 4.82%, ThermoMPNN-D 5.27%, and Kermut 5.29%. StabilityArc + Kermut reaches 8.12%. Mean MLP and mean ridge reach 7.30% and 7.48%. The advantage extends to combinations whose measured endpoints are separated from both assay limits. Figure 4a,b displays the common-main-effect and margin comparisons.

5.5 Reference constraints improve residual prediction

The four-state contrast improves mean residual accuracy against the same-capacity direct head. The direct head's residual RMSE is 0.635 kcal/mol under the primary grouping and 0.630 kcal/mol under the stricter grouping. SCEPI's corresponding errors are 0.595 and 0.604 kcal/mol. Its native hit-rate gain is 0.52 percentage points under the primary grouping and −0.23 points under the stricter grouping. The contrast supplies exact reference constraints and a residual-prediction benefit across both groupings. Tables 8 and 9 report the complete prediction and selection results.

Removing the learned pair term at inference retains the trained single head and phenotype map. Native hit rate falls from 11.22% to 4.42%, and residual RMSE rises from 0.595 to 1.072 kcal/mol. This intervention establishes the pair output's contribution within the trained predictor (Figure 4c,d). Representation, mapping, and training-objective variants are reported with their complete results in the appendix.

6 Discussion

SCEPI transfers an interaction function conditioned on measured single effects. Matched controls establish the additional prediction and selection value beyond external calibration under both homology groupings. This input mechanism complements StabilityArc's cross-protein prior for target Gaussian-process adaptation [13]. The full-target comparison demonstrates its use when target single measurements are available and combination labels remain withheld.

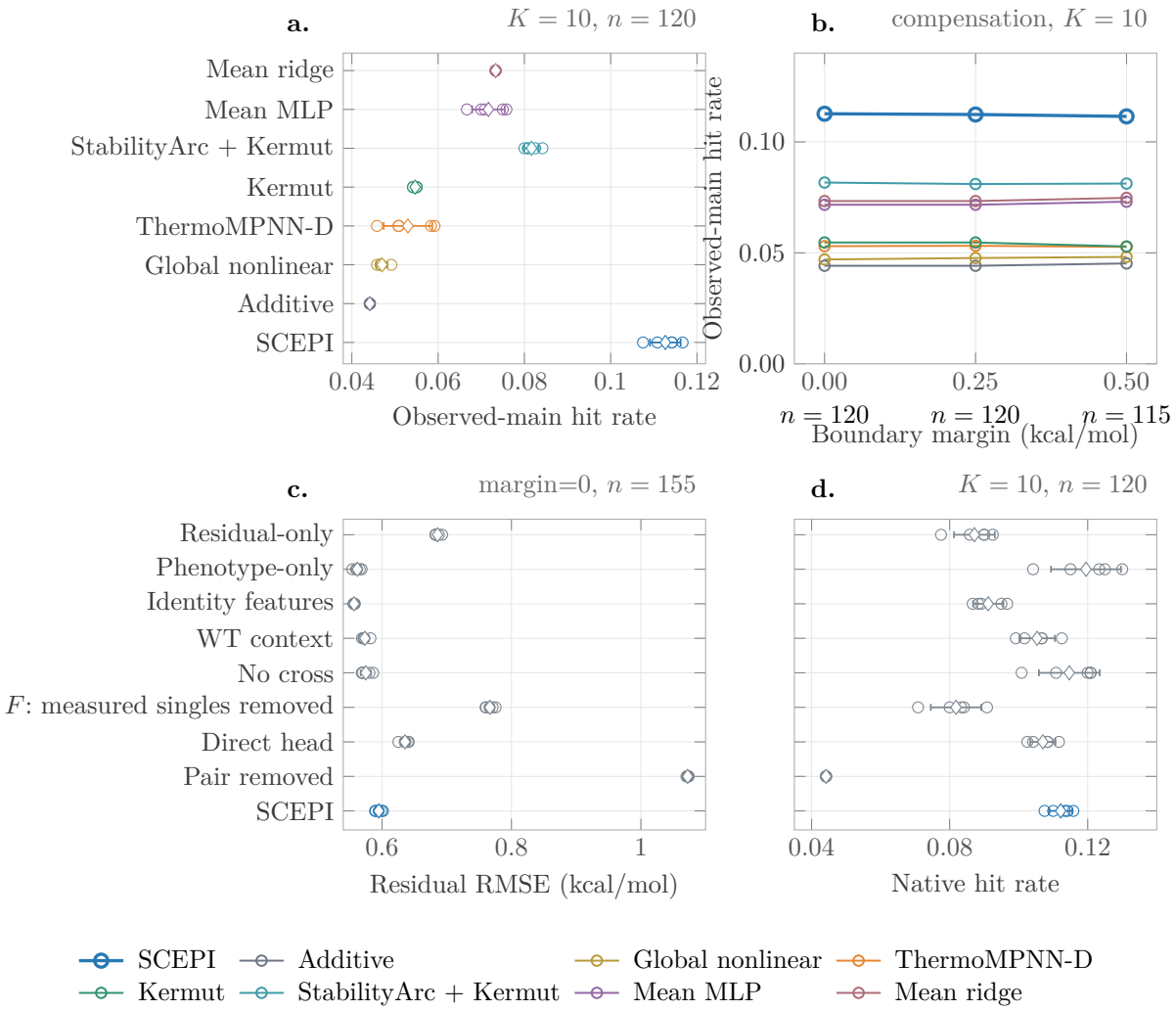


Fig. 4. Measured-single conditioning and component comparisons under 30% identity grouping. (a) Top-10 hit rates with common measured main effects. (b) Quartet boundary-margin sensitivity; cohort sizes appear below the axis. (c) Residual RMSE and (d) native hit rate share the labeled control order. The direct head has matched capacity. The input-removal control retains external calibration and removes measured effects and availability from F. Open points are equal-protein repetition means; diamonds and whiskers show their mean and standard deviation. Residual and standard selection cohorts contain 155 and 120 held-out proteins.

The four-state construction supplies exact reference constraints and lower mean residual error than a matched direct head. Its compensation hit-rate differences vary across groupings. Functional and higher-order tests evaluate a separate two-stage variant with predicted single effects. Their heterogeneous gains define the task-dependent performance of target adaptation.

7 Conclusion

SCEPI conditions a transferable interaction function on measured single effects. Matched controls and held-out comparisons establish gains in residual prediction and compensatory discovery. Code, configurations, and reproduction materials are available at `https://github.com/SZU-ADDG/SCEPI`.

References

[1] N. Azbukina, A. Zharikova, and V. Ramensky. Intragenic compensation through the lens of deep mutational scanning. *Biophysical Reviews*, 14(5):1161–1182, 2022. doi:10.1007/s12551-022-01005-w.

[2] B. H. Davis, A. F. Y. Poon, and M. C. Whitlock. Compensatory mutations are repeatable and clustered within proteins. *Proceedings of the Royal Society B*, 276(1663):1823–1827, 2009. doi:10.1098/rspb.2008.1846.

[3] H. Dieckhaus and B. Kuhlman. Protein stability models fail to capture epistatic interactions of double point mutations. *Protein Science*, 34(1):e70003, 2025. doi:10.1002/pro.70003.

[4] A. J. Faure and B. Lehner. MoCHI: neural networks to fit interpretable models and quantify energies, energetic couplings, epistasis, and allostery from deep mutational scanning data. *Genome Biology*, 25:303, 2024. doi:10.1186/s13059-024-03444-y.

[5] L. I. Gong, M. A. Suchard, and J. D. Bloom. Stability-mediated epistasis constrains the evolution of an influenza protein. *eLife*, 2:e00631, 2013. doi:10.7554/eLife.00631.

[6] P. M. Groth, M. H. Kerrn, L. Olsen, J. Salomon, and W. Boomsma. Kermut: Composite kernel regression for protein variant effects. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi:10.52202/079017-0929.

[7] Z. Lin et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637):1123–1130, 2023. doi:10.1126/science.ade2574.

[8] J. Ouyang-Zhang, D. Diaz, A. Klivans, and P. Krahenbuehl. Predicting a protein's stability under a million mutations. In *Advances in Neural Information Processing Systems*, volume 36, 2023. doi:10.52202/075280-3332.

[9] J. Otwinowski, D. M. McCandlish, and J. B. Plotkin. Inferring the shape of global epistasis. *Proceedings of the National Academy of Sciences*, 115(32):E7550–E7558, 2018. doi:10.1073/pnas.1804015115.

[10] A. Tareen, M. Kooshkbaghi, A. Posfai, et al. MAVE-NN: learning genotype-phenotype maps from multiplex assays of variant effect. *Genome Biology*, 23:98, 2022. doi:10.1186/s13059-022-02661-7.

[11] S. Reeves and S. Kalyaanamoorthy. Robust multi-mutant protein stability prediction from a fine-tuned evolutionary scale model. bioRxiv preprint, 2026. doi:10.64898/2026.06.04.730231.

[12] K. Tsuboyama et al. Mega-scale experimental analysis of protein folding stability in biology and design. *Nature*, 620:434–444, 2023. doi:10.1038/s41586-023-06328-6.

[13] A. L. Feller, A. D. Ellington, and C. O. Wilke. StabilityArc: Decoding protein sequence embeddings into generalizable stability landscapes. arXiv preprint arXiv:2610.00742, 2026.

[14] Y. Park, B. P. H. Metzger, and J. W. Thornton. The simplicity of protein sequence-function relationships. *Nature Communications*, 15:7953, 2024. doi:10.1038/s41467-024-51895-5.

[15] P. Notin et al. ProteinGym: Large-Scale Benchmarks for Protein Fitness Prediction and Design. In *Advances in Neural Information Processing Systems*, volume 36, pages 64331–64379, 2023.

[16] K. Didi et al. FLIP2: Expanding Protein Fitness Landscape Benchmarks for Real-World Machine Learning Applications. In *Proceedings of the 43rd International Conference on Machine Learning*, 2026.